

Waaberi Ibrahim

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EDUCATION

University of Ottawa

Ottawa, ON, Canada

BASc in Software Engineering, Co-op

GPA: 3.9/4.0 | Expected Graduation: Dec 2029

Activities: Webmaster, IEEE uOttawa; VP of Development, uOttawa Cybersecurity Club; Software Developer, uOttHack; Director, Hack the Hill

EXPERIENCE

Software Engineer Intern

May 2026 – Present

Statistics Canada

Ottawa, ON

- Built a **Python** tool converting **7 legacy e-courses** to editable web lessons; saved **hundreds of hours** of manual work
- Enabled **12 employees** to find answers to training questions by deploying a **RAG assistant** using approved docs
- Reached **130+ employees** via the **Gatsby/React** AI Beginner Program site; earned an award for my contribution
- Built an **R** library applying Statistics Canada publication standards to any visualization, automating styling across reports

Undergraduate Research Assistant, Health Devices

May 2026 – Present

Computational Analysis and Acceleration Research Group, University of Ottawa

Ottawa, ON

- Built a **React/Hono** operator console for contactless vital-sign and fall monitoring, streaming real-time alerts via **SSE**
- Engineered an **MQTT** pipeline to validate and deduplicate **3 telemetry streams** before writing to **PostgreSQL**
- Built server-side **RBAC** for **17 permissions/5 roles**, gating alert closure on protocol completion and medical sign-off
- Implemented a **DevOps GitOps** workflow for commit-triggered, health-checked deployments of a **5-service Docker stack**

Webmaster (Previously Software Developer & Technical Coordinator)

Jan. 2026 – Present

IEEE uOttawa Student Branch

Ottawa, ON

- Own and deploy a live **Next.js/Payload CMS** site serving **4,000+ visits** and **20,000+ impressions** monthly
- Shipped bilingual routing across **20+ CMS fields**, static generation, sitemaps, meta tags, and automated backups
- Contributed **23 merged PRs** and **73 commits** as top contributor in an **8-contributor** open-source repository
- Reached **50 participants** by delivering a web security workshop on common vulnerabilities in CTFs

PROJECTS

Sentinel | *FastAPI, React, XGBoost, Docker* | [Demo](#) | [GitHub](#)

- Built a containerized **FastAPI** backend that ingests CSV/XLSX trade histories and serves session-scoped analysis
- Trained an **XGBoost** model for bias scoring, recommendations, and what-if simulation with **91% accuracy**
- Implemented **8 REST endpoints** for metrics, bias scoring, P/L analysis, recommendations, and what-if simulation
- Published a working **YouTube demo**; delivered with a **4-person team** and won **3rd Place** at the hackathon

Quark | *QNX, C++, TensorFlow Lite, Raspberry Pi* | [Devpost](#) | [GitHub](#)

- Built a **QNX driver-safety system** that detects drowsiness from camera input and triggers audio alerts
- Ran vision and language inference fully on-device with **TensorFlow Lite** and **llama.cpp**, avoiding cloud dependencies
- Won **3rd Place in the QNX Challenge at cuHacking7** with a **3-person team** and working embedded-AI prototype

Synacor VM Emulator | *Python, Virtual Machines, Low-Level Systems* | [GitHub](#)

- Implemented a working **22-opcode**, 15-bit Synacor virtual machine in **Python**
- Built an 8-register CPU with **32,768-word memory**, stack operations, control flow, and terminal I/O
- Added an interactive **debugger** with stack inspection and runtime analysis to solve the challenge

SKILLS

Languages: Java, Python, JavaScript, TypeScript, R, C#, Bash, Go, Lua, Ruby, SQL, C++

Frameworks: React, Next.js, Node.js, FastAPI, Django

ML & Data: PyTorch, TensorFlow, llama.cpp, XGBoost, scikit-learn, Qiskit, Polars, Pandas, NumPy

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLite

Tools: Git, Docker, Linux, Hugging Face, GitHub Copilot, Google Colab, Grafana, GitHub Actions, Azure, JUnit, PyTest, gdb

AWARDS AND CERTIFICATIONS

3rd Place - cuHacking7 QNX Challenge	2026
1st Place - IEEE uOttawa x HackTheBox Capture The Flag	2026
3rd Place - uOttawa SESA National Bank Hackathon	2026
1st Place - IBM Qiskit Fall Fest 2025 Hackathon (Quantum Computing)	2025
1st Place - uOttawa-IBM Cyber Range FlagForce Capture The Flag	2025
2nd Place - University of Ottawa Computer Chess Hackathon	2025